

AI Driven Privacy Preserving Digital Ecosystem for Modernizing Sri Lanka's Paddy Supply Chain

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Project Proposal Report

Ransara N.S. – IT22102096

B.Sc. (Hons) Degree in Information Technology Specialized in
Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Ransara N.S.	IT22102096	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Supervisor:

Name	Date	Signature
Dr. Mahima Weerasinghe	2026/03/16	

Co-Supervisor:

Name	Date	Signature
Mr. Eishan Weerasinghe	2026/03/14	

ABSTRACT

Paddy cultivation plays a vital role in Sri Lanka's agricultural economy and food security. However, in recent years the country has experienced a gradual decline in paddy farmers due to unstable market prices and reduced profitability. Price fluctuations during harvest seasons make it difficult for farmers to estimate future income, discouraging them from continuing paddy cultivation. As a result, domestic rice production decreases, forcing the government to import rice from foreign markets, which negatively impacts the national economy. Although the government aims to strengthen rice self-sufficiency and promote rice exports, the lack of reliable price forecasting and decision-support tools limits farmers' ability to plan their cultivation effectively.

Existing studies on paddy or rice price forecasting primarily rely on traditional statistical models such as ARIMA or basic machine learning approaches like linear regression, neural networks, and LSTM. These approaches mainly focus on predicting prices using historical data but do not provide explanations for price fluctuations or integrate policy-related knowledge that influences market behavior. Furthermore, limited research has explored advanced transformer-based forecasting models or causal reasoning mechanisms for agricultural price analysis, particularly in the Sri Lankan context.

To address these limitations, this research proposes an AI-Driven Paddy Price Forecasting System that integrates Transformer-based time-series models, Causal Machine Learning, and Large Language Model (LLM) reasoning. The system utilizes multiple data sources including historical paddy prices, weather and climate data, production volume statistics, government policy information, market demand indicators, and seasonal cultivation patterns. Transformer architectures such as Informer or FEDformer will be used for long-range price forecasting, while causal machine learning techniques will identify key factors influencing price fluctuations. Additionally, a Retrieval-Augmented Generation (RAG) framework with LLMs will provide human-readable explanations referencing relevant government policies and market conditions.

The expected outcome is an intelligent decision-support system that not only predicts future paddy prices but also explains the underlying causes of price changes. This system can help farmers make informed cultivation decisions, support policymakers in evaluating market interventions, and contribute to stabilizing Sri Lanka's agricultural economy. Socially, the system can encourage farmer retention and sustainable agricultural practices, while commercially it can improve supply chain planning and reduce reliance on rice imports.

Keywords: Paddy Price Forecasting, Transformer Time-Series Models, Causal Machine Learning, Agricultural AI, Explainable AI.

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LIST OF ABBREVIATIONS

Abbreviation	Full Term
AI	Artificial Intelligence
ANN	Artificial Neural Network
ARIMA	Autoregressive Integrated Moving Average
API	Application Programming Interface
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
DL	Deep Learning
ML	Machine Learning
MLP	Multi-Layer Perceptron
LSTM	Long Short-Term Memory
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
EMD	Empirical Mode Decomposition
IoT	Internet of Things
JSON	JavaScript Object Notation
GPU	Graphics Processing Unit
API	Application Programming Interface
PMB	Paddy Marketing Board
WBS	Work Breakdown Structure
NLP	Natural Language Processing

INTRODUCTION

1.1 Background and Context

Agriculture remains one of the most critical sectors in Sri Lanka's economy, contributing significantly to national food security, rural employment, and economic stability. Among the various agricultural commodities, paddy (rice) is the most important staple crop in Sri Lanka, forming the primary food source for the majority of the population. Paddy cultivation has historically played a central role in the country's socio-economic and cultural structure, supporting thousands of farming families and contributing substantially to rural livelihoods. Sri Lanka cultivates paddy mainly during two major agricultural seasons known as the Maha and Yala seasons, which together account for the majority of the country's domestic rice production. According to data published by the Department of Census and Statistics and the Department of Agriculture, rice consumption in Sri Lanka exceeds 100 kg per person annually, making rice the dominant staple food in the country.

Despite the historical importance of paddy cultivation, Sri Lanka is currently facing a gradual decline in the number of active paddy farmers. One of the major reasons behind this issue is the instability and unpredictability of paddy market prices. Farmers often begin cultivation without having reliable information about the price they will receive at the time of harvest, which typically occurs several months after planting. As a result, farmers frequently face situations where the market price during the harvest period is significantly lower than expected, leading to financial losses. This uncertainty discourages farmers from continuing paddy cultivation and has contributed to a steady reduction in the farming population. When domestic production decreases due to reduced farming activity, the country is often forced to import rice from foreign markets to meet local demand. Such imports place additional pressure on national foreign reserves and negatively affect the country's economy.

The issue of paddy price fluctuations is influenced by multiple interconnected factors. Climatic conditions such as excessive rainfall, droughts, and floods can significantly impact crop yields and harvest quantities. When production is reduced due to adverse weather conditions, market supply decreases and prices tend to rise. Conversely, favorable weather conditions and high production levels may result in an oversupply of paddy, causing prices to drop during harvest seasons. In addition to climatic factors, government policies such as guaranteed price schemes, import restrictions, and market regulations also influence price behavior. Furthermore, demand for rice, population growth, and seasonal cultivation patterns contribute to the dynamic nature of the paddy market. Because of these complex interactions, predicting paddy prices accurately remains a challenging task.

Several studies have attempted to address agricultural price forecasting using statistical and machine learning methods. Traditional forecasting approaches such as the Autoregressive Integrated Moving Average (ARIMA) model have been widely used in agricultural price prediction studies. For example, multiple studies conducted in India have used ARIMA models to forecast paddy prices using historical market data. While these methods provide reasonable short-term forecasting capabilities, they are limited in their ability to capture complex nonlinear relationships between multiple influencing factors such as climate conditions, production volume, and market demand. Other research has explored machine learning models including Artificial Neural Networks (ANN), Linear Regression, Random Forest, and Long Short-Term Memory (LSTM) networks for price forecasting. Although these approaches improve prediction performance compared to traditional statistical models, most of them focus primarily on historical price data and do not incorporate broader contextual factors such as government policy changes or supply chain dynamics.

Another important limitation of existing solutions is the lack of interpretability and explanation for price fluctuations. Many predictive models generate numerical forecasts but fail to provide meaningful explanations regarding the reasons behind predicted price changes. For farmers and policymakers, understanding why prices are expected to increase or decrease is equally important as the prediction itself. Additionally, existing research rarely integrates official policy documents, agricultural regulations, or institutional knowledge into the forecasting process. This results in prediction systems that are technically accurate but lack contextual awareness of real-world agricultural policies and market regulations.

To address these limitations, this research proposes an AI-Driven Paddy Price Forecasting System that integrates advanced artificial intelligence techniques to provide both accurate predictions and meaningful explanations. The proposed solution incorporates several key components. First, transformer-based time series forecasting models such as Informer or FEDformer will be utilized to analyze long-term historical data and predict future paddy prices. Transformer architectures have recently demonstrated strong performance in long-range time series forecasting due to their ability to capture complex temporal dependencies in large datasets. Second, causal machine learning techniques will be applied to identify and analyze the relationships between various influencing factors such as weather conditions, production volumes, seasonal patterns, and market demand. This approach enables the system to provide explanations for predicted price changes by identifying causal relationships within the data. Finally, a Retrieval-Augmented Generation (RAG) framework combined with a Large Language Model (LLM) will be used to generate human-readable explanations for price forecasts by referencing government policy documents and institutional reports related to paddy marketing and agricultural regulation.

The proposed research is also aligned with several United Nations Sustainable Development Goals (SDGs). In particular, it contributes to SDG 2: Zero Hunger, which focuses on ensuring sustainable food production systems and improving agricultural productivity. By providing farmers with reliable price forecasts and decision-support tools, the system can help reduce financial risks and encourage continued participation in agricultural activities. Additionally, the research supports SDG 8: Decent Work and Economic Growth, as stabilizing the agricultural sector can enhance rural employment opportunities and strengthen the national economy. The research also indirectly contributes to SDG 9: Industry, Innovation, and Infrastructure by introducing advanced artificial intelligence technologies into the agricultural domain.

From an academic and technological perspective, this research aligns with the broader research cluster of Artificial Intelligence for Sustainable Agriculture and Smart Food Systems. This cluster focuses on applying modern data science, machine learning, and AI technologies to address challenges in agricultural production, food security, and supply chain management. By combining transformer-based forecasting models, causal machine learning methods, and explainable AI techniques, the proposed system represents an interdisciplinary approach that bridges agricultural economics, artificial intelligence, and data-driven policy analysis.

Overall, the proposed research aims to develop an intelligent decision-support system capable of forecasting future paddy prices while simultaneously explaining the factors that drive market fluctuations. Such a system has the potential to support farmers in making informed cultivation decisions, assist policymakers in evaluating market interventions, and contribute to improving the stability and sustainability of Sri Lanka's paddy farming sector.

1.2 Literature Review

Agricultural commodity price forecasting has become an important research area due to its significant impact on farmers' income, food security, and market stability. Price fluctuations in agricultural markets are influenced by multiple factors such as weather conditions, seasonal production, government policies, and supply–demand dynamics. Therefore, accurate forecasting models are essential to help farmers, policymakers, and agricultural stakeholders make informed decisions. Over the past two decades, researchers have explored various statistical, machine learning, and deep learning techniques to predict agricultural commodity prices, including paddy and rice.

Statistical Approaches for Agricultural Price Forecasting

Early studies on agricultural price prediction mainly relied on traditional statistical time-series models such as the Autoregressive Integrated Moving Average (ARIMA) model. ARIMA models analyze historical price patterns and use them to forecast future values. Several studies conducted in agricultural markets have demonstrated that ARIMA can effectively model linear trends and seasonal patterns in commodity prices. However, these models are limited in capturing complex nonlinear relationships between multiple influencing factors. As a result, their performance often decreases when market conditions become highly volatile or when multiple external variables affect price behavior.

For example, many studies on rice and agricultural commodity price forecasting in countries such as India and Indonesia have applied ARIMA models using historical market price data. While these models provide reasonable short-term forecasts, they typically rely only on historical price series and do not incorporate additional variables such as climate conditions or production volumes. This restricts their ability to reflect real-world market dynamics.

Machine Learning-Based Approaches

With the advancement of data analytics and computing capabilities, machine learning models have increasingly been applied to agricultural price prediction. Techniques such as Linear Regression, Support Vector Regression (SVR), Random Forest, and XGBoost have been used to analyze price trends and identify relationships between multiple variables. Machine learning methods are capable of handling large datasets and identifying nonlinear patterns that cannot be captured by traditional statistical models.

Recent studies have shown that ensemble learning methods can significantly improve the accuracy of agricultural price forecasting. For example, ensemble regression models combining rainfall data and wholesale price indices have been used to predict agricultural commodity price trends more accurately than individual regression models.

However, machine learning models often require extensive feature engineering and careful selection of input variables. In many cases, these models still depend heavily on historical price data, and they rarely provide interpretable explanations regarding the causes of price changes.

Deep Learning Approaches

Deep learning techniques have recently gained attention for time-series forecasting tasks due to their ability to model complex temporal dependencies. Models such as Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) networks have been widely used for agricultural price forecasting. These models can capture nonlinear patterns and long-term dependencies in time-series data, making them suitable for predicting volatile commodity prices.

Research comparing ARIMA and LSTM models for agricultural price prediction has shown that deep learning approaches can outperform traditional statistical models in certain cases because they can learn complex patterns from large datasets.

In addition, hybrid approaches that combine statistical and deep learning models have been proposed to improve forecasting performance. For example, hybrid ARIMA–LSTM models integrate statistical modeling with neural network learning to capture both linear and nonlinear patterns in agricultural price data. Such hybrid models have demonstrated improvements in prediction accuracy compared with standalone statistical models.

Despite these improvements, deep learning models often function as “black boxes,” meaning that they generate predictions without providing clear explanations for the underlying causes of price fluctuations. This lack of interpretability makes it difficult for farmers and policymakers to trust and adopt these systems in practical decision-making.

Existing Systems and Applications

Several digital platforms and agricultural information systems have been developed to provide farmers with market information and price updates. These systems typically offer historical price data, daily market prices, and simple trend analysis tools. In some cases, web-based applications have been developed to automate agricultural commodity price prediction using machine learning algorithms such as ARIMA, Prophet, or LSTM.

Although these systems can provide basic forecasting capabilities, most existing platforms focus primarily on numerical prediction. They rarely integrate additional contextual information such as government policies, climate data, or supply chain dynamics. Furthermore, most systems do not offer explanation mechanisms that help users understand why prices are predicted to increase or decrease.

Limitations of Existing Research and Systems

Although significant progress has been made in agricultural price forecasting, several limitations remain in the existing research literature and practical systems.

First, many existing studies rely heavily on historical price data without incorporating broader economic, environmental, and policy-related factors that influence agricultural markets. This limits the ability of forecasting models to capture real-world market dynamics.

Second, most research focuses primarily on prediction accuracy while neglecting interpretability and explainability. Farmers and policymakers require explanations of price fluctuations to make informed decisions, but current models rarely provide such insights.

Third, many existing models are designed for short-term forecasting and struggle to capture long-term dependencies in time-series data. Recent studies suggest that newer time-series foundation models and advanced architectures may significantly improve forecasting performance compared with traditional approaches.

Finally, there is limited research focusing specifically on the Sri Lankan agricultural market, particularly in the context of paddy price forecasting. Most existing studies have been conducted in other countries such as India, Malaysia, and Bangladesh, which may have different agricultural systems, policies, and market dynamics.

Summary

In summary, previous studies have explored various techniques for agricultural price forecasting, ranging from traditional statistical models to advanced machine learning and deep learning approaches. While these methods have improved prediction accuracy, they still suffer from several limitations, including limited interpretability, insufficient integration of external factors, and a lack of context-specific research for Sri Lanka. These limitations highlight the need for a more advanced and integrated approach that combines accurate forecasting with explainable insights into the factors influencing paddy price fluctuations.

1.3 Research Gap

Although numerous studies have explored agricultural commodity price forecasting, several important gaps remain in the existing literature and practical systems. Identifying these gaps is essential in order to justify the need for a new research approach that can better address the challenges faced by paddy farmers and agricultural policymakers.

One of the most significant limitations in existing research is the heavy reliance on traditional statistical forecasting methods, particularly the Autoregressive Integrated Moving Average (ARIMA) model. Many studies conducted in countries such as India and Bangladesh have successfully applied ARIMA models to forecast paddy or rice prices using historical time-series data. While ARIMA is effective in modeling linear patterns and short-term seasonal trends, it struggles to capture complex nonlinear relationships and long-range dependencies that often exist in agricultural markets. Paddy price behavior is influenced by multiple interacting variables including climate conditions, production volume, seasonal cycles, government policies, and market demand. Traditional statistical models are not designed to incorporate and learn from such complex multidimensional data, which limits their forecasting accuracy and adaptability in dynamic market environments.

In recent years, machine learning and deep learning techniques such as Artificial Neural Networks (ANN), Linear Regression, Random Forest, and Long Short-Term Memory (LSTM) networks have been introduced to improve agricultural price prediction. These approaches can model nonlinear relationships and analyze larger datasets compared to traditional statistical models. However, most of these models still focus primarily on historical price trends and do not adequately incorporate broader contextual information such as government policy interventions, supply chain disruptions, or climate variability. Furthermore, many machine learning models require extensive feature engineering and may produce predictions without clearly explaining the factors responsible for price fluctuations.

Another critical limitation of existing approaches is the lack of interpretability and explainability. Most predictive models are designed to generate numerical price forecasts but do not provide meaningful explanations regarding why the price is expected to increase or decrease. For farmers, traders, and policymakers, understanding the underlying reasons behind price changes is essential for effective decision-making. Without explanations, predictive systems may appear as “black-box” solutions, which reduces trust and practical adoption in real-world agricultural environments.

Additionally, current research rarely integrates policy knowledge or institutional information into price forecasting systems. Government policies such as guaranteed price schemes, subsidies, import restrictions, and market regulations significantly influence agricultural markets. However, most existing forecasting models ignore these policy factors and rely solely on numerical datasets. As a result, these

systems lack contextual awareness and cannot provide policy-informed insights that are useful for decision makers.

Another important gap is the limited application of advanced transformer-based time series models in agricultural price forecasting. Transformer architectures such as Informer and FEDformer have recently demonstrated strong performance in long-range time series forecasting tasks due to their ability to capture complex temporal dependencies across large datasets. Despite their proven effectiveness in other domains such as energy forecasting and financial prediction, these models have not yet been widely applied to paddy price forecasting, particularly within the context of developing agricultural economies.

Furthermore, there is very limited research focusing specifically on Sri Lanka's paddy market. Most existing studies have been conducted in countries such as India, Malaysia, Indonesia, and Bangladesh. While these studies provide useful insights into agricultural price forecasting techniques, the agricultural ecosystems, climate conditions, government policies, and market structures in those countries differ significantly from those in Sri Lanka. Therefore, applying models developed in other regions may not accurately capture the unique characteristics of the Sri Lankan paddy market.

Considering these limitations, there is a clear need for a more advanced and integrated approach to paddy price forecasting. The proposed research addresses this gap by introducing a novel AI-driven framework that combines transformer-based time series forecasting, causal machine learning analysis, and large language model reasoning. Unlike existing systems that focus solely on numerical prediction, the proposed approach aims to provide both accurate price forecasts and explainable insights into the factors influencing price fluctuations. Transformer-based models will enable improved long-range forecasting performance, while causal machine learning methods will help identify relationships between variables such as climate conditions, production levels, and market demand. Additionally, a Retrieval-Augmented Generation (RAG) framework integrated with a large language model will allow the system to generate human-readable explanations by referencing relevant government policies and agricultural reports.

This integrated approach represents a significant advancement over existing research by combining prediction, explanation, and policy-aware reasoning within a unified system. By addressing the limitations of current forecasting methods and focusing specifically on the Sri Lankan agricultural context, this research contributes both methodological innovation and practical value to the field of agricultural data analytics. The proposed system has the potential to support farmers in making informed cultivation decisions, assist policymakers in evaluating market interventions, and ultimately contribute to improving the stability and sustainability of the paddy farming sector in Sri Lanka.

To identify the research gap, we compared seven existing research in more recent works to our solution. The results are shown in Table.

Feature / Capability	Research [1]	Research [2]	Research [3]	Research [4]	Research [5]	Research [6]	Research [7]	Proposed System
Uses Traditional Statistical Models (ARIMA etc.)	Yes	Yes	Yes	Yes	No	No	Yes	No
Uses Machine Learning / Deep Learning Models	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Uses Transformer-based Time Series Models	No	No	No	No	No	No	No	Yes
Incorporates Climate / Weather Factors	No	No	Yes	No	No	Yes	No	Yes
Incorporates Production Volume Data	No	No	Yes	No	No	Yes	No	Yes
Integrates Government Policy Information	No	No	No	No	No	No	No	Yes
Provides Explainable / Interpretable Predictions	No	No	Yes (LIME)	No	No	No	No	Yes (Causal ML + LLM)
Uses Causal Machine Learning Analysis	No	No	No	No	No	No	No	Yes
Provides Human-Readable Explanations	No	No	Limited	No	No	No	No	Yes (LLM + RAG)
Focused on Sri Lankan Paddy Market	No	No	No	No	No	No	No	Yes
Unified Framework (Forecast + Explain + Policy Reasoning)	No	No	No	No	No	No	No	Yes

Table 1 Comparative Analysis of Existing Paddy Price Forecasting Studies and the Proposed System

1.4 Research problem

Paddy cultivation plays a critical role in Sri Lanka's food security and agricultural economy. However, in recent years the country has experienced a gradual decline in the number of paddy farmers due to unstable market prices and reduced profitability. Farmers often begin cultivation without reliable information about the future price of paddy at the harvest period, which typically occurs several months after planting. This uncertainty exposes farmers to significant financial risks because the market price during harvest can fluctuate due to various factors such as weather conditions, production volume, seasonal supply, government policy changes, and market demand.

As a result, many farmers face situations where the selling price of paddy is lower than their production cost, leading to financial losses and discouraging them from continuing agricultural activities. The reduction in the farming population subsequently leads to decreased domestic rice production. When local supply becomes insufficient to meet national demand, Sri Lanka is forced to import rice from foreign markets, which places additional pressure on national foreign reserves and negatively affects the country's economy.

Although several forecasting techniques have been proposed in previous research to predict agricultural commodity prices, most existing systems rely on traditional statistical models or basic machine learning approaches that primarily analyze historical price data. These models often fail to capture complex relationships between multiple influencing factors such as climate variability, seasonal production patterns, government price control policies, and market demand. Furthermore, many existing systems focus only on generating price predictions without providing clear explanations about the causes of price fluctuations, which limits their usefulness for farmers and policymakers who require interpretable insights for decision-making.

Another limitation is the absence of intelligent systems that integrate policy knowledge and contextual information related to agricultural markets. Government policies such as guaranteed price schemes, import restrictions, and agricultural subsidies play a major role in influencing paddy prices. However, existing forecasting approaches rarely incorporate such information into their analytical models.

Therefore, the main research problem addressed in this study is the lack of an accurate, explainable, and policy-aware paddy price forecasting system that can help farmers and policymakers understand and anticipate future market price changes in the Sri Lankan paddy sector.

To address this problem, there is a need for an advanced AI-driven framework that combines modern time-series forecasting models, causal analysis techniques, and intelligent reasoning mechanisms to both predict future paddy prices and explain the factors influencing price fluctuations.

OBJECTIVE

2.1 Main Objective

To develop an AI-driven paddy price forecasting system that accurately predicts future paddy prices and explains price fluctuations using transformer-based time-series models, causal machine learning analysis, and policy-aware reasoning techniques.

2.2 Specific Objectives

Objective 1

To collect, preprocess, and integrate multi-source agricultural datasets including historical paddy price data, weather and climate data, production volume statistics, seasonal cultivation patterns, and market demand indicators relevant to the Sri Lankan paddy market.

Objective 2

To develop a transformer-based time-series forecasting model (such as Informer or FEDformer) capable of predicting future paddy prices using historical price data and multiple influencing factors.

Objective 3

To implement causal machine learning techniques that identify and analyze relationships between key variables such as weather conditions, production levels, seasonal patterns, and market demand that influence paddy price fluctuations.

Objective 4

To design a Retrieval-Augmented Generation (RAG) framework integrated with a Large Language Model (LLM) that generates human-readable explanations for predicted price changes by referencing relevant government policies and agricultural reports.

Objective 5

To develop an intelligent decision-support system that integrates price prediction, causal analysis, and explainable reasoning to assist farmers and policymakers in understanding and responding to paddy market dynamics.

Objective 6

To evaluate the performance of the proposed forecasting system using standard prediction accuracy metrics and interpretability analysis to ensure its effectiveness in real-world agricultural decision-making.

METHODOLOGY

This research adopts a System Development and Evaluation methodology combined with an Experimental research approach to design, implement, and validate an AI-driven paddy price forecasting system. The methodology is selected because the primary goal of the research is to develop a novel intelligent software system and demonstrate measurable improvements in forecasting accuracy and explainability compared to existing approaches.

In the context of Software System Technologies (SST), the research focuses on designing a data-driven software architecture that integrates machine learning models, causal reasoning techniques, and intelligent explanation mechanisms to support decision-making in the agricultural sector. The proposed system will be developed as an intelligent decision-support platform capable of forecasting paddy prices and explaining the factors that influence market fluctuations.

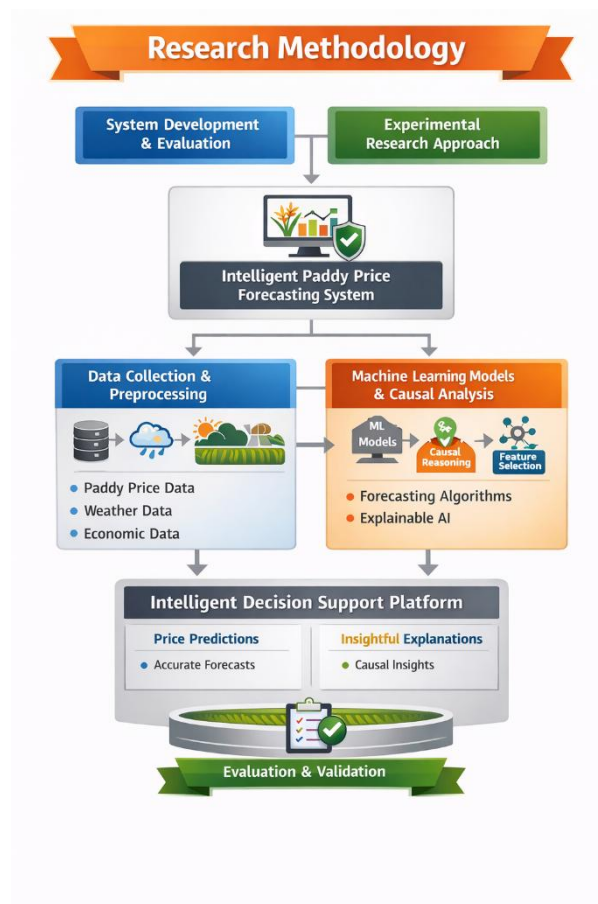


Figure 1 Research Methodology diagram

3.1 Research Approach

The study follows a System Development and Evaluation methodology, which is widely used in Software Engineering research when developing a new system or improving an existing technological solution. The proposed system will be designed, implemented, and evaluated to determine whether it provides measurable improvements in forecasting performance and decision-support capabilities.

The research process consists of several phases. First, relevant datasets related to paddy prices, weather conditions, production volumes, seasonal cultivation patterns, and market demand will be collected and integrated into a unified dataset. Second, the system architecture will be designed to support data processing, forecasting, causal analysis, and explanation generation. Third, machine learning models will be implemented and trained using historical data. Finally, the proposed system will be evaluated through controlled experiments and compared with baseline forecasting methods.

In addition, the research incorporates an experimental evaluation phase where multiple forecasting models will be tested under identical conditions to measure prediction accuracy and system performance. This ensures that the results are supported by quantitative and reproducible evidence, which is a key requirement in empirical software engineering research.

3.2 Technical Methods and Algorithms

The proposed system integrates several advanced artificial intelligence techniques to perform accurate forecasting and generate interpretable insights.

The main forecasting component uses Transformer-based time-series models, such as Informer or FEDformer, which are designed for long-range time series forecasting. These models use attention mechanisms to capture complex temporal relationships within large datasets and have demonstrated superior performance compared to traditional statistical and recurrent neural network models.

To analyze the factors influencing price fluctuations, Causal Machine Learning techniques will be applied. These techniques identify relationships between variables such as rainfall, production volume, seasonal cultivation patterns, and market demand. Causal analysis enables the system to explain how changes in these variables contribute to predicted price movements.

In addition, the system incorporates Large Language Model (LLM) reasoning combined with a Retrieval-Augmented Generation (RAG) framework. This module retrieves relevant policy documents, agricultural reports, and institutional data from sources such as the Paddy Marketing Board and the Central Bank of Sri Lanka. The LLM then generates human-readable explanations describing why specific price changes are predicted.

The complete system workflow includes the following stages:

1. Data collection and preprocessing
2. Feature engineering and dataset integration
3. Transformer-based price forecasting
4. Causal relationship analysis
5. Policy-aware explanation generation
6. System output generation for decision support

3.3 Tools and Platforms

The proposed system will be implemented using modern software development and machine learning tools.

The primary programming language used for development will be Python, which provides extensive libraries for data analysis and artificial intelligence applications. Machine learning and deep learning models will be implemented using frameworks such as PyTorch or TensorFlow, which provide efficient support for training transformer-based architectures.

Data preprocessing and statistical analysis will be performed using Pandas, NumPy, and Scikit-learn, which are widely used in machine learning workflows. Causal analysis techniques may be implemented using specialized libraries such as DoWhy or EconML, which support causal inference in data-driven applications.

For the explanation module, natural language processing frameworks combined with Retrieval-Augmented Generation (RAG) techniques will be used to integrate Large Language Models with domain-specific knowledge sources.

Development and experimentation will be conducted using environments such as Jupyter Notebook, Google Colab, or GPU-enabled computing platforms, which allow efficient training and testing of deep learning models.

These tools are selected because they provide scalability, strong community support, and efficient integration capabilities required for building advanced AI-driven software systems.

3.4 Data Collection and Ethical Considerations

The proposed research will utilize multiple datasets obtained from reliable government and institutional sources related to the Sri Lankan agricultural sector.

Historical paddy price data will be obtained from institutions such as the Paddy Marketing Board and the Department of Census and Statistics. Weather and climate data will be collected from the Department of Meteorology, while agricultural production statistics will be obtained from the Department of Agriculture.

Additional economic indicators related to rice demand and market behavior will be obtained from publications and reports provided by the Central Bank of Sri Lanka. Seasonal cultivation information related to Maha and Yala seasons will also be incorporated into the dataset to capture seasonal market patterns.

Since the research primarily uses publicly available or institutional datasets and does not involve personal or sensitive information, the ethical risks associated with data usage are minimal. However, all datasets will be used in compliance with research ethics guidelines, and proper attribution will be provided for all data sources. Data integrity and transparency will also be maintained during preprocessing and model training stages.

3.5 Validation Strategy and Performance Metrics

The proposed system will be evaluated using a structured validation strategy to ensure reliability and scientific rigor.

The collected dataset will be divided into training, validation, and testing subsets to evaluate the forecasting model on unseen data. Cross-validation techniques may also be applied to improve model generalization and reduce overfitting.

The performance of the proposed transformer-based forecasting model will be compared with baseline models such as ARIMA, Linear Regression, and LSTM networks. This baseline comparison is essential to demonstrate the effectiveness of the proposed system.

Forecasting accuracy will be evaluated using commonly used time-series performance metrics, including:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)

These metrics provide quantitative evidence of the model's prediction accuracy and enable direct comparison with existing forecasting techniques.

In addition to prediction accuracy, the interpretability and reasoning capabilities of the system will also be evaluated. The causal analysis component will be assessed based on its ability to identify meaningful relationships between influencing variables, while the explanation module will be evaluated based on the relevance and clarity of generated explanations.

Through this comprehensive evaluation process, the research aims to demonstrate that the proposed system provides improved forecasting performance, better interpretability, and enhanced decision-support capabilities compared to existing paddy price forecasting approaches.

HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed solution introduces an AI-driven, privacy-preserving digital ecosystem designed to modernize Sri Lanka's paddy supply chain by integrating intelligent analytics, secure data management, and automated decision-support technologies. The system addresses several structural challenges in the current agricultural ecosystem, including lack of predictive insights, poor coordination across stakeholders, absence of real-time monitoring, and limited trust in data sharing.

The architecture consists of four major AI-driven components that collaboratively process agricultural, economic, and supply-chain data to generate actionable insights for farmers, policymakers, warehouse operators, and market participants.

The first component is a Digital Twin for Paddy Farming, which creates a continuously updated virtual model of paddy fields by integrating multimodal data such as satellite images, climate data, soil measurements, and farm metadata. Deep learning vision models and graph neural networks enable crop health monitoring and environmental interaction modelling.

The second component, AI-Driven Paddy Price Forecasting, focuses on predicting long-term paddy price trends using Transformer-based time-series models combined with causal machine learning. This component provides transparent explanations of price changes by analyzing economic indicators, climate factors, and production data. It directly supports farmers and policymakers in making informed market decisions.

The third component is a Secure Disaster-Aware Paddy Warehouse Coordination System, which ensures transparent and reliable warehouse management using blockchain technology and privacy-preserving cryptographic protocols. This component improves trust and traceability in stock reporting and logistics operations.

The fourth component is an AI-Powered Farmer-Miller Marketplace, which enables secure and intelligent trading interactions between farmers and rice millers. Federated learning ensures that user data remains private while AI-driven negotiation agents help optimize transaction outcomes.

Together, these components form an integrated ecosystem that combines AI intelligence, secure data sharing, and intelligent decision support to improve efficiency, transparency, and sustainability within the paddy supply chain.

4.2 System Diagram of Components

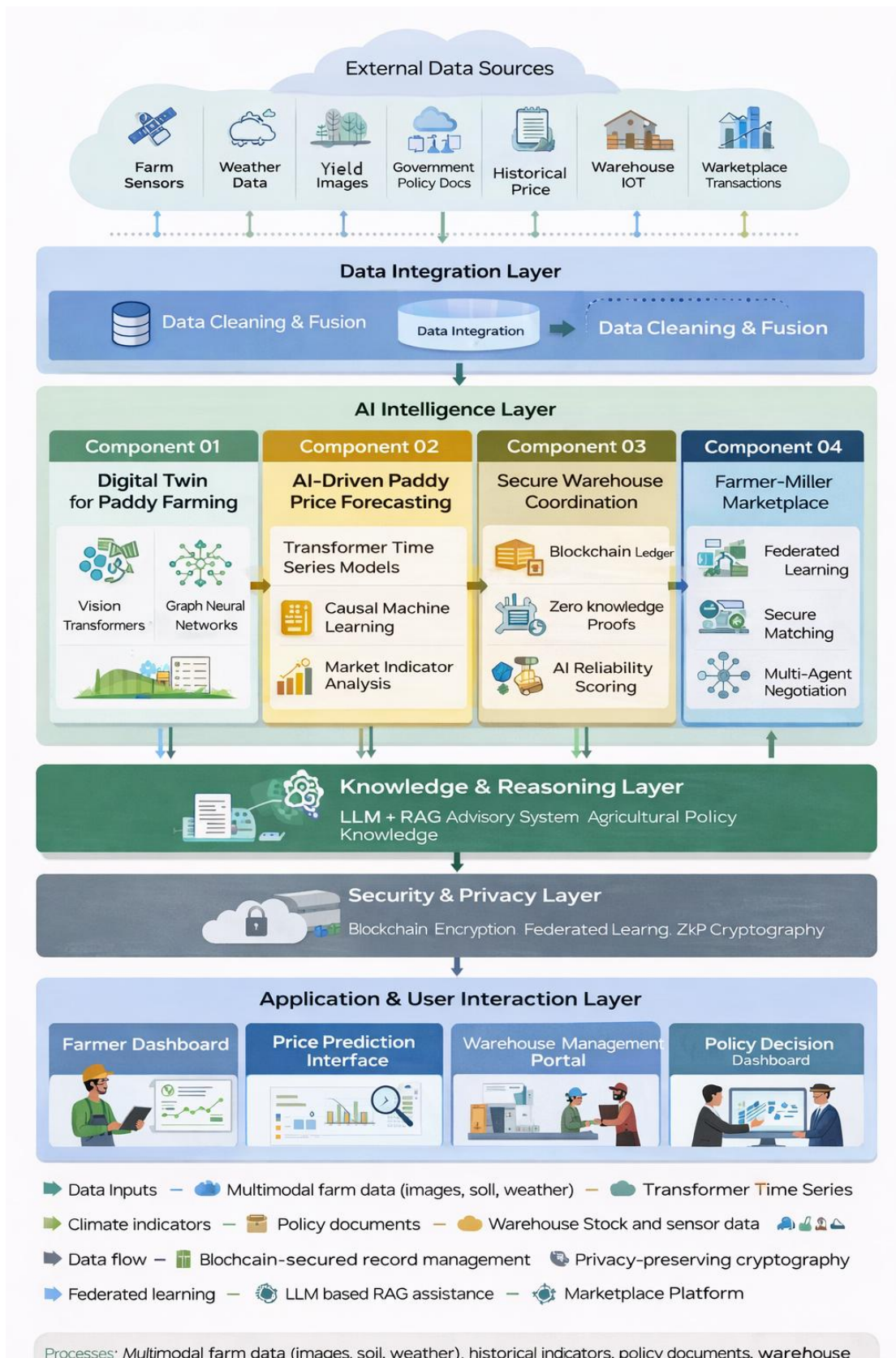


Figure 2 System Diagram of Component

4.3 Individual Component Highlight – AI-Driven Paddy Price Forecasting

The AI-Driven Paddy Price Forecasting component plays a central role in providing predictive market intelligence within the ecosystem.

This component is designed to overcome the limitations of traditional statistical forecasting methods by integrating Transformer-based time-series models, causal machine learning, and LLM-based reasoning. Transformer architectures such as Informer or FEDformer are used to capture long-range dependencies within historical price data, enabling accurate forecasting of future price trends.

The component also incorporates causal machine learning techniques to identify relationships between price fluctuations and influencing factors such as climate variations, production levels, government policies, and market demand. This allows the system not only to predict prices but also to explain the underlying causes behind market changes.

The internal modules of this component include:

- Data preprocessing and feature extraction for historical price and climate datasets
- Transformer-based time-series forecasting model
- Causal inference module for identifying key drivers of price fluctuations
- LLM reasoning module for generating explainable insights
- Price prediction API service for integration with other system components

The output of this component provides future price predictions, causal explanations, and market intelligence insights that support decision-making across the supply chain.

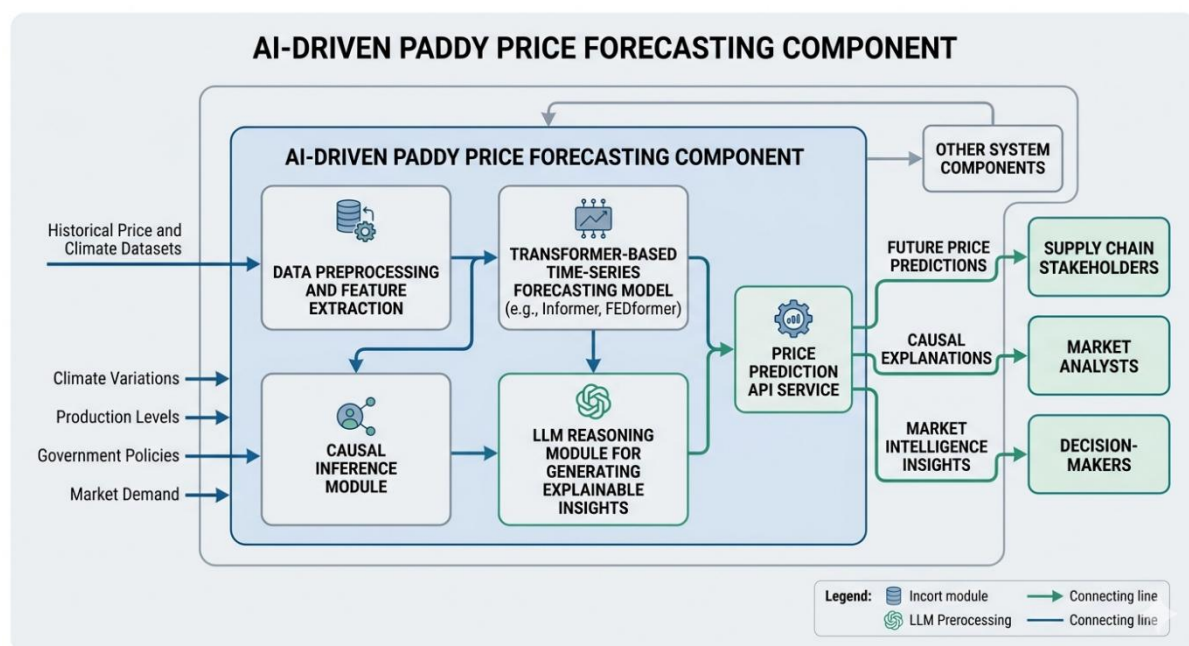


Figure 3 System Diagram for the AI Driven Paddy Price Forecasting Component

4.4 Component Interactions

The AI-Driven Paddy Price Forecasting component operates as an independent analytical module within the overall agricultural intelligence ecosystem. Its primary function is to analyze historical and external market-related datasets and generate future paddy price predictions.

Unlike other system components, the Price Forecasting module does not directly interact with the Digital Twin, Warehouse Coordination, or Marketplace components during its core analytical process. Instead, it relies on external datasets and publicly available information sources, such as historical paddy price records, weather conditions, market demand indicators, and government policy data.

The processed data is passed through the forecasting pipeline, where machine learning and deep learning models analyze patterns and trends to produce price predictions. These predictions are then delivered to stakeholders through the application interface, enabling informed decision-making.

Although the component functions independently at the system level, its outputs can be consumed by external systems or other modules in the future if integration is required. This modular architecture ensures that the forecasting service remains scalable, maintainable, and adaptable to future system expansion.

4.5 Data Flow Explanation

The AI-Driven Paddy Price Forecasting component processes agricultural and market data through a structured pipeline to generate accurate paddy price predictions and explain the factors influencing price fluctuations.

Initially, raw data is collected from multiple external sources, including historical paddy price databases, climate and weather datasets, government policy announcements, and market transaction records. These datasets provide the fundamental indicators required to analyze market trends and supply–demand dynamics.

The collected data enters the Data Integration and Preprocessing Layer, where several operations are performed to prepare the data for analysis. These operations include data cleaning, normalization, handling missing values, and feature extraction. Relevant features such as seasonal patterns, rainfall levels, production estimates, and historical price trends are extracted to improve model performance.

After preprocessing, the prepared dataset is forwarded to the AI Forecasting Layer. In this stage, machine learning and deep learning models analyze temporal patterns and relationships between influencing factors to generate future paddy price predictions. These models learn from historical price behavior and external variables to estimate short-term and long-term market price movements.

To improve transparency and trust in the predictions, the system also includes an Explanation and Reasoning Layer. In this stage, explainable AI techniques analyze the model outputs and identify the key factors responsible for price fluctuations, such as weather changes, supply shortages, policy changes, or seasonal demand variations. The system then generates human-readable explanations describing why the predicted price changes may occur.

Finally, the generated price forecasts and explanatory insights are delivered through the Application Layer, where users such as farmers, policymakers, and market analysts can access the results. This allows stakeholders to understand future price trends and the reasons behind them, supporting more informed agricultural and market-related decision-making.

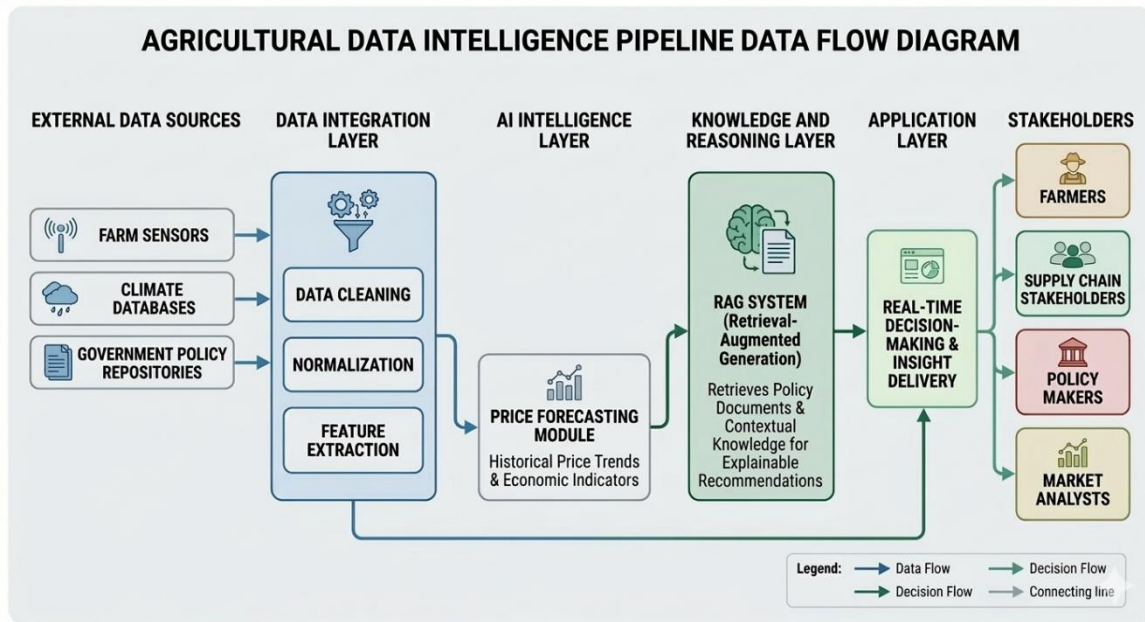


Figure 4 Data Flow Diagram

4.6 Scalability and Security Considerations

The architecture is designed to support scalability and secure data management across a large agricultural ecosystem. A modular microservice-based architecture allows each component to operate independently and scale according to workload demands. Cloud-based infrastructure can be used to dynamically allocate computational resources for machine learning training, forecasting tasks, and real-time data processing.

Security and privacy are addressed through multiple mechanisms. Blockchain technology ensures immutable storage of warehouse transactions, preventing unauthorized data manipulation. Privacy-preserving cryptographic techniques such as zero-knowledge proofs protect sensitive operational data while still allowing verification of records.

Additionally, federated learning enables collaborative machine learning without requiring centralized storage of private user data. This ensures that farmers and marketplace participants retain control over their data while still benefiting from collective intelligence generated by the system.

These architectural considerations ensure that the proposed ecosystem remains scalable, secure, and reliable, making it suitable for large-scale deployment within Sri Lanka's agricultural sector.

USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed AI-Driven Paddy Price Forecasting system is designed to support several stakeholders involved in the Sri Lankan paddy supply chain. These stakeholders rely on accurate price predictions and market insights to make informed decisions regarding production, storage, trading, and policy formulation.

The primary stakeholder of the system is the paddy farmer, who directly benefits from early market intelligence. Farmers often make cultivation, harvesting, and selling decisions without reliable future price information. By providing accurate price forecasts and explanations of market trends, the system enables farmers to select optimal selling periods and reduce financial uncertainty.

The secondary stakeholders include government agencies, rice millers, and agricultural market participants who use price predictions for planning and operational decision-making.

Key stakeholders include:

Primary Stakeholder

- Farmers – Use the system to obtain predicted paddy prices and market insights that support decisions related to harvesting, storage, and selling.

Secondary Stakeholders

- Paddy Marketing Board (PMB) – Uses price forecasting insights to plan purchasing policies, regulate market prices, and maintain stable rice supply.
- Rice Millers – Use predicted price trends to plan raw material procurement and milling operations.
- Agricultural Marketers / Traders – Use forecasting insights to analyze market trends and make strategic trading decisions.
- Agricultural Policymakers and Researchers – Use forecasting analytics to monitor market stability and evaluate agricultural policy impacts.

These stakeholders interact with the system through dashboards, reporting tools, and predictive analytics interfaces that provide accessible and understandable forecasting insights.

5.2 Functional Requirements

The functional requirements describe the core functionalities that the AI-Driven Paddy Price Forecasting system must provide.

FR1 – Data Collection

The system shall collect historical paddy price data from relevant agricultural databases and institutional sources.

FR2 – External Data Integration

The system shall integrate external influencing factors such as weather conditions, climate indicators, production statistics, and seasonal cultivation data.

FR3 – Data Preprocessing

The system shall preprocess collected datasets through data cleaning, normalization, and feature extraction to prepare them for machine learning analysis.

FR4 – Price Forecast Generation

The system shall generate future paddy price forecasts using Transformer-based time-series models.

FR5 – Causal Factor Identification

The system shall identify key variables influencing price changes using causal machine learning techniques.

FR6 – Explanation Generation

The system shall generate human-readable explanations describing the factors that influence predicted price changes.

FR7 – Price Prediction Dashboard

The system shall provide a visual dashboard displaying forecasted prices, historical trends, and analytical insights.

FR8 – Forecast Visualization

The system shall present predicted price trends using charts and visual analytics to improve user understanding.

FR9 – Forecast Query Interface

The system shall allow users to request predictions for specific time horizons (e.g., weekly, monthly, or seasonal forecasts).

FR10 – Data Update Mechanism

The system shall periodically update forecasting models when new data becomes available.

FR11 – API Service for Predictions

The system shall provide an API service that allows other systems or applications to retrieve price prediction results.

5.3 Non-Functional Requirements

Non-functional requirements describe the performance, security, usability, and operational characteristics of the system.

Security Requirements

NFR1 – The system shall ensure secure access to forecasting services through authentication mechanisms.

NFR2 – Sensitive agricultural and institutional datasets shall be protected from unauthorized access.

NFR3 – Data integrity shall be maintained to prevent manipulation of price forecasting results.

Performance Requirements

NFR4 – The system shall generate price predictions within an acceptable response time (e.g., under 5 seconds for forecast queries).

NFR5 – The system shall support large historical datasets without significant performance degradation.

Scalability Requirements

NFR6 – The system shall support increasing volumes of agricultural data and users through scalable infrastructure.

Usability Requirements

NFR7 – The forecasting dashboard shall provide simple visualizations that are easy for non-technical users such as farmers to understand.

NFR8 – The system interface shall support clear graphical representation of price trends and predictions.

Reliability Requirements

NFR9 – The system shall ensure consistent prediction availability even when data updates occur.

NFR10 – The system shall maintain stable operation during large data processing tasks.

Maintainability Requirements

NFR11 – The system shall allow model updates and improvements without requiring major system redesign.

NFR12 – The system shall support integration with additional datasets or analytical modules in future system upgrades.

COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed AI-Driven Paddy Price Forecasting system primarily targets stakeholders involved in Sri Lanka's agricultural market, particularly those connected to the paddy supply chain. The system is designed to support decision-making by providing accurate market predictions and explainable insights.

The primary target users are paddy farmers, who require reliable information about future price trends in order to determine the best time to sell their harvest. Many farmers currently rely on informal market information or middlemen, which often leads to financial disadvantages.

The secondary target users include government institutions such as the Paddy Marketing Board, agricultural policymakers, rice millers, and agricultural traders. These stakeholders use price forecasts to plan procurement strategies, market regulation policies, and storage operations.

A typical customer persona for the system is a small or medium-scale paddy farmer who needs early market intelligence to reduce price uncertainty and maximize profit. Another persona includes policymakers and agricultural analysts who require reliable forecasting tools to monitor market stability and make data-driven decisions.

Although the platform will be released publicly as a free national agricultural service, the main "customer" of the developed system is the Sri Lankan government or agricultural authorities, who can deploy the solution to support farmers and strengthen the national food supply chain.

6.2 Value Proposition

The proposed system provides significant value by introducing data-driven intelligence into Sri Lanka's paddy market, which currently lacks accurate and transparent price forecasting mechanisms.

The key value proposition of the system includes:

- Accurate price forecasting using advanced AI models that analyze historical market data and external economic factors.
- Explainable insights that help users understand why price changes occur rather than simply showing predictions.
- Decision-support for farmers, enabling them to plan harvesting, storage, and selling strategies more effectively.
- Market stability support for policymakers by providing predictive insights into price fluctuations.
- Accessible and free agricultural intelligence service that can benefit farmers across the country.

By improving the availability of reliable market information, the system helps reduce exploitation by middlemen and supports fairer agricultural markets.

6.3 Revenue Model

Although the system will be made available free of charge to farmers and public users, the development and deployment of the platform can still follow a sustainable financial model through institutional support.

The most appropriate revenue or sustainability model is a government-supported digital public infrastructure model. In this approach, the system is developed through academic research and technology innovation, and later deployed by government agencies such as the Paddy Marketing Board or Ministry of Agriculture.

Potential financial support sources include:

- Government agricultural digitalization programs
- International agricultural development grants
- Public–private partnerships with agricultural technology providers
- Research and innovation funding initiatives

The economic value generated by the system is not direct user revenue but indirect national economic benefits, such as improved farmer income, better market efficiency, and reduced price volatility.

6.4 Cost Estimation

The development cost of the proposed system is relatively low due to the use of open-source technologies, free development tools, and publicly available datasets.

Major cost factors include:

Development Costs

- AI model development and experimentation
- Data preprocessing and integration
- System interface and dashboard development

Infrastructure Costs

- Cloud computing resources for model training and deployment
- Data storage and system hosting

Operational Costs

- Periodic model retraining using updated datasets
- System maintenance and performance monitoring

Because the system relies on free datasets and open-source frameworks such as Python, TensorFlow/PyTorch, and data science libraries, the overall financial investment required for development is significantly reduced.

6.5 Pricing Strategy

The proposed solution follows a public-service pricing strategy, where the system is made freely accessible to farmers and agricultural stakeholders.

Instead of charging individual users, the system can be maintained through government funding or institutional partnerships, ensuring that all farmers regardless of financial capacity can access the forecasting service.

This approach supports digital inclusivity in agriculture and ensures that technological benefits are distributed across rural communities.

6.6 Competitive Advantage

The proposed system introduces several technological and practical advantages compared to existing agricultural information platforms.

First, the system uses Transformer-based time-series models, which can capture long-range dependencies in market data and provide more accurate long-term price forecasts than traditional statistical models.

Second, the integration of causal machine learning techniques allows the system to identify the key factors influencing price changes. This makes the system more transparent and informative compared to conventional prediction systems that operate as black-box models.

Third, the system incorporates LLM-based reasoning and explanation generation, enabling the platform to provide clear, human-readable insights about market conditions.

Finally, the platform is designed specifically for the Sri Lankan agricultural context, using local datasets, seasonal patterns, and policy information. This localized focus provides more relevant insights than generic agricultural forecasting tools.

6.7 Intellectual Property Considerations

The intellectual property of the developed system may be managed through a collaborative ownership model involving the research institution and relevant government authorities.

Key intellectual property considerations include:

- The AI forecasting algorithms and system architecture may be documented as part of academic research contributions.
- The software platform and trained models may be transferred to the government for public deployment.
- Open-source software frameworks used during development will comply with their respective licensing agreements.

Since the primary objective of the system is public benefit and national economic improvement, the platform may be released under a public-access or open innovation model, allowing government agencies to maintain and expand the system for long-term use.

BUDGET AND JUSTIFICATION

The proposed AI-Driven Paddy Price Forecasting System is developed as an undergraduate research component using existing personal hardware resources, open-source software tools, and free-tier cloud services. As a result, the implementation of this component does not require direct financial investment at this stage. The development approach prioritizes cost efficiency while ensuring the system can still be implemented and evaluated effectively.

Budget Justification Summary

The proposed component is designed to be cost-efficient and accessible, leveraging open-source technologies, free datasets, and free-tier cloud services. Because the researcher already possesses the required development hardware and the project is conducted as part of academic research, no additional financial investment is required at this stage.

This zero-cost development model demonstrates the feasibility of implementing advanced AI-based forecasting solutions even with limited resources. Furthermore, once deployed at a national level, the system could be integrated into existing government digital infrastructure to support farmers and agricultural stakeholders without imposing financial barriers on users.

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WORK BREAKDOWN STRUCTURE (WBS)

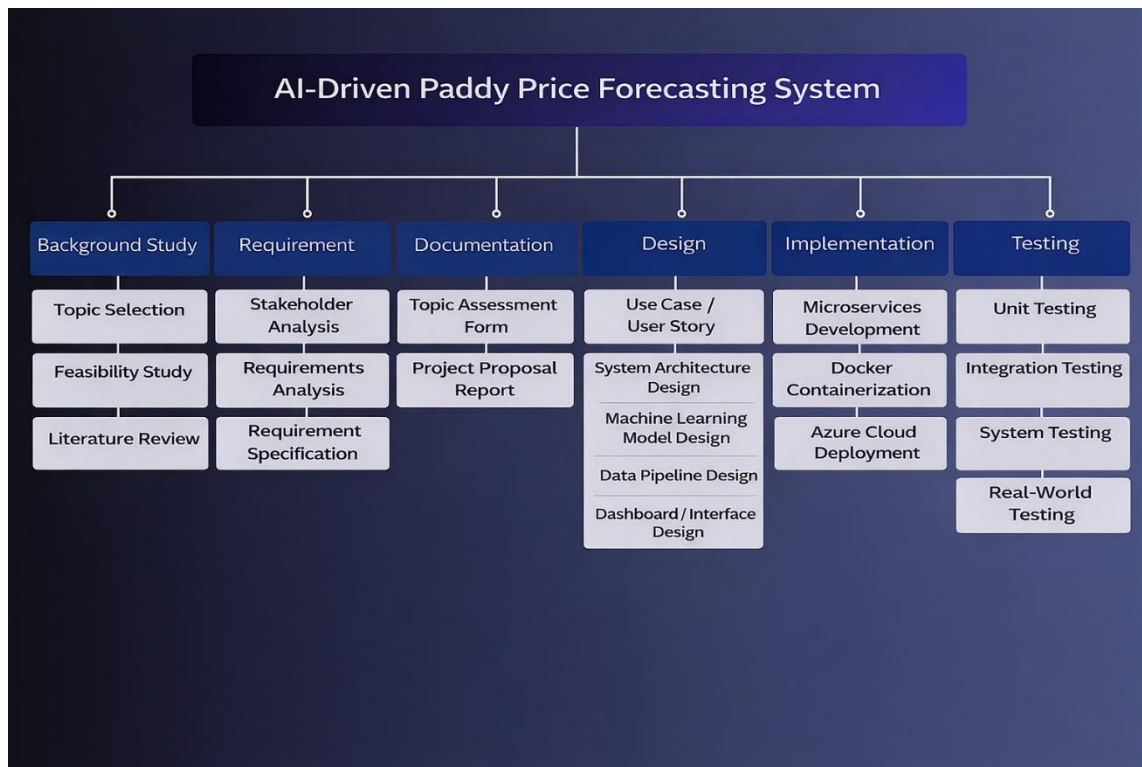


Figure 5 Work Breakdown Structure

GANTT CHART

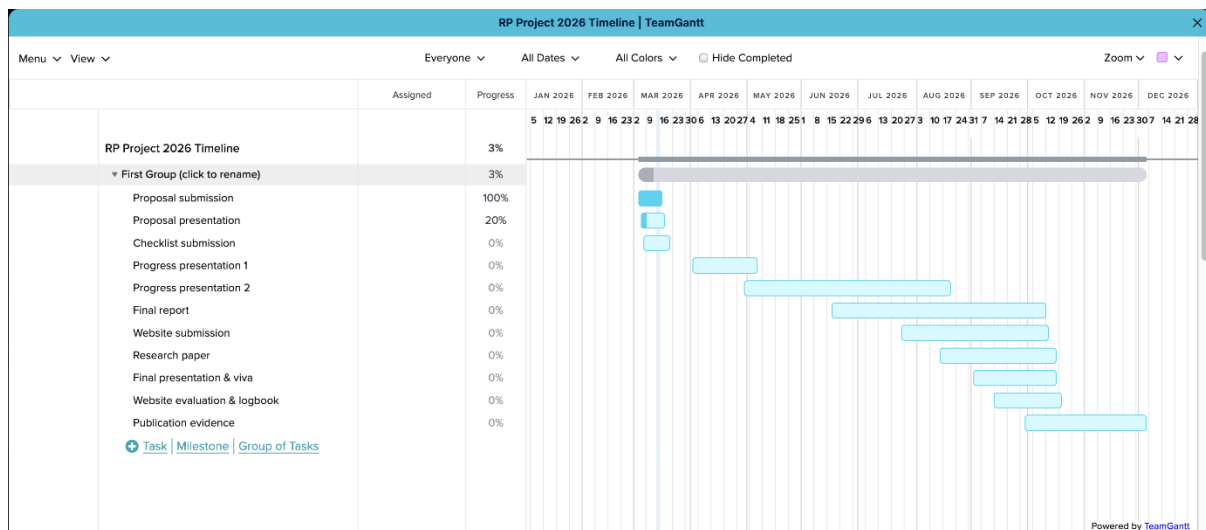


Figure 6 Grantt Chart

APPENDIX

Gantt Chart Trello Board : <https://trello.com/b/ZMqS4l0B/rp-project-2026-timeline>